MIL-T-10540D
8 November 1978
SUPERSEDING
MIL-T-10540C
6 August 1969

MILITARY SPECIFICATION

TRACING TABLE, STEREOPLOTTER, PROJECTION:

WITH CASE

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a multiplex stereoplotter tracing table with case.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
GGG-C-111	- Calipers and Gages, Vernier.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	- Preservation-Packaging, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-N-3497	- Nadirscope, Stereoplotter Tracing Table.
MIL-C-12044	- Chests, Plywood.
MIL-I-18422	- Indicators, Dial and Accessories.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MS15536	- Lamp, Incandescent, T-3-1/4, Miniature Screw Base.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C33.1 - Flexible Cord and Fixture Wire.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Description. The projection stereoplotter tracing table with case (hereinafter called "tracing table") shall consist of a base, vertical standards, platen, elevating screw, pencil attachment, counter, and a case.

3.2 First article (first produced tracing table with case). The contractor shall furnish one or more tracing tables as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove that his production methods and choice of design detail will produce tracing tables that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.6).

3.4 Environmental conditions.

3.4.1 Storage. The tracing table shall be capable of storage under the following conditions when tested as specified in 4.5.2.1.1 and 4.5.2.1.2.

Air temperature:

High - plus 160° F (plus 71° C), time period, 1 hour.
Low - minus 30° F (minus 34° C), time period, 5 hours.
No solar radiation.

3.5 Weight and dimensions. The tracing table shall not exceed 260 millimeters (mm) (10.236 inches) in overall height and shall not weigh (excluding the case) more than 5 pounds (2.27 kg).

3.6 Base. The base of the tracing table shall be a lightweight, metal alloy, U-shaped casting. Space shall be provided in the base for installation of the necessary wiring, switches, and lamps specified herein. The base shall be provided with three adjustable foot supports. The foot supports shall be fabricated of quartz or agate. That portion of the foot support which rests on the plotting table shall be ground flat, shall be between 0.250 inch (6.35 mm) and 0.375 inch (9.525 mm) in diameter, and shall be highly polished.

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3.7 Vertical standards. The vertical standards shall be straight and rigidly mounted on the base parallel to each other and perpendicular to the plane of the foot supports. The standards shall be constructed to permit attachment of a nadirscope conforming to MIL-N-3497. The vertical standards shall be of sufficient height to permit at least 100 mm vertical travel of the table platen (see 3.9).

3.8 Platen. The platen shall be a flat disk approximately 110 mm in diameter, the surface of which will be finished in a flat white. The platen shall be equipped with a cone-shaped sapphire lens built into its center. The tip of the cone shall project through the surface of the platen to form a semispherical point approximately 0.1 mm in diameter. The base of the cone shall be exposed to an aperture through which light from beneath the platen may enter the sapphire lens. The illuminated tip of the lens shall form a "floating mark" visible without loss of clarity at all angles of incidence from 0 to 75 degrees. The platen shall be held firmly in a mounting without rocking upon its support, and shall rotate through 360 degrees without horizontal displacement of the sapphire lens point in the center of the platen more than 0.1 mm in any direction. The quality of the surface coating of the platen shall be such that it will withstand repeated cleaning with soap and warm water without damage.

3.9 Platen mount. The platen mount shall support the platen in a horizontal plane and shall contain adjustable ball-bearing rollers fitted against the vertical standards (see 3.7) to guide the platen in a vertical motion throughout the range of the tracing table. The rollers shall be positioned to cause minimum interference with the hand of the operator. The platen mount shall house the lamp provided for the illumination of the sapphire lens in the center of the platen.

3.10 Platen extension. The platen extension shall raise the platen 75 mm, plus or minus 0.25 mm, above its normal position on the platen mount. The platen extension shall rotate on the platen mount through 360 degrees without horizontal displacement of the sapphire lens point in the center of the platen more than 0.1 mm in any direction.

3.11 Elevating screw and lock. The vertical movement of the platen mount shall be effected by a right-hand, double-threaded Acme screw, 10 mm outside diameter and 5 mm lead. A coarse-knurled knob 50 mm in diameter shall be provided for the operation of the vertical screw.

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The operator shall be able to turn the knurled knob with either hand when his hands are placed on the base of the instrument in position for drawing. The vertical motion shall operate so that the operator can raise and lower the platen with the motion of one finger. There shall be no play in this motion. The nut in the platen mount bearing on the vertical screw shall prevent backlash and yet allow for takeup in case of wear. The platen mount shall be capable of being locked in any position throughout its range of travel.

3.12 Pencil-holder attachment. The pencil-holder attachment shall be mounted on the base directly beneath the tracing table platen. The attachment shall hold graphite pencil leads 1.9 to 2.0 mm in diameter and at least 20 mm in length. The pencil point shall be capable of being raised from and lowered to the drawing surface beneath the platen without damage to the lead point. The position of the graphite lead shall be adjustable horizontally so that it may be centered vertically beneath the sapphire lens point in the center of the platen. The horizontal adjustments shall be made by adjusting pins. Locks shall be provided on the horizontal adjustments so that the setting will not be disturbed in normal operation of the tracing table or when graphite leads are changed. The weight of the pencil-holder attachment shall be 2-3/4 ounces (77.96 grams). Attachable weights of 1/2, 1, and 1-1/2 ounces (14.18, 28.35, and 42.53 grams) shall be furnished to be placed on the pencil-holder attachment to increase the friction between the pencil point and the drawing paper.

3.13 Counter. A counter shall be provided on the tracing table for indicating the displacement of the platen in a vertical direction. The counter shall be mounted at the top of the left-hand vertical standard so that it may be read without lifting the table. The counter shall read in millimeters directly to 0.1 mm and shall be arranged so that interpolation may be made to the nearest 0.05 mm. The counter shall cover the full range of the tracing table platen from zero to 100 mm, and shall indicate the true vertical displacement of the tracing table platen throughout the full range within 0.05 mm. The counter shall be capable of being set to any desired reading within 0.05 mm without altering the present height of the platen. The counter shall have a white index mark. The whole number digits shall be white with a black background. The tenths digit shall be black with a white background.

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3.14 Illumination and wiring. Electric lamps shall be provided in the tracing table as follows: One lamp for illumination of the sapphire lens in the tracing table platen both with and without the platen extension; one lamp for illumination of the counter; and two lamps for illumination of the pencil point and the immediately surrounding area on the map sheet. All lamps shall be 6- to 8-volts, 0.25-ampere, miniature screw base, T-3-1/4, incandescent lamps conforming to MS15536. The lamp for illumination of the sapphire lens in the tracing table platen shall be controlled by a rheostat with "off" position. The rheostat shall be placed in the base of the tracing table and shall be operated by a small, flat, knurled knob that will not interfere with the use of the tracing table. Provision shall be made for centering this lamp to obtain maximum illumination. A resistor shall be wired in series with this lamp to reduce the 20 volts supplied by the transformer to 6 volts. The lamps for illumination of the pencil point and the counter shall be wired in series and shall be controlled by an "on-off" micro-switch mounted in the base of the instrument convenient to the operator. All wiring and cord shall conform to ANSI C33.1, Type TP. All lamps shall be shielded from direct view by the operator in any working position. The tracing table shall be equipped with a 5-foot (1.524 m) flexible cord with standard two-wire, parallel blade, male connector.

3.15 Case. A case shall be provided to house the tracing table. Partitions and padded blocks shall be provided for cushioning the components within the case. Material, construction, painting, marking, workmanship, and hardware of the case shall conform to MIL-C-12044 and the following:

1/2-inch plywood.
2 handles.
2 drawbolts.
2 hinges.
1 cover chain.
1 loading plan.
Color when specified (see 6.2).
Marking:

U. S.

Tracing Table,

Stereoplotter

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3.16 Identification marking. The tracing table shall be identified in accordance with MIL-STD-130.

3.17 Treatment and painting. The portions of the tracing table normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A. Color, unless otherwise specified (see 6.2), shall be at the option of the contractor.

3.18 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.19 Workmanship. All parts and components of the tracing table shall be clean and free from sand, dirt, fins, sprues, scale, flux, and other harmful extraneous material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First produced tracing table inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

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4.3 First produced tracing table inspection.

4.3.1 Examination. The first produced tracing table shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first produced tracing table shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, Inspection Level II.

4.4.2 Examination.

4.4.2.1 Samples. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 1.5 percent defective for major defects and 6.5 percent defective for minor defects.

4.4.3 Tests.

4.4.3.1 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 1.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The tracing tables shall be examined as specified herein for the following defects:

Major

101. Material and weight not as specified.
102. Base not as specified.
103. Vertical standards will not accommodate the nadirscope, not perpendicular to the base, or not parallel to each other.
104. Dimensions not as specified.
105. No provisions for locking platen in position.
106. Low diffusion or reflection on platen.
107. Pencil attachment will not hold specified leads.

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108. Locks not provided for pencil attachment.
109. Counter location and range not as specified.
110. Setting counter disturbs platen setting.
111. Illumination and wiring provisions not as specified.
112. Sapphire lens missing or does not project through the surface of the platen.
113. Floating mark not clearly visible at all angles of incidence specified.
114. Lamp controls do not operate as required.
115. Play in elevating screw.
116. Platen extension not as specified.
117. Case not as specified.
118. Interchangeability of parts not as specified.
119. Workmanship not as specified.

Minor

201. Identification marking not as specified.
202. Treatment and painting not as specified.

4.5.2 Tests.

4.5.2.1 Storage temperatures.

4.5.2.1.1 Low temperature. The case with tracing table, repair parts, and accessories within the case, shall be placed in a test chamber at 72° F (22 °C). Temperature in the chamber shall be lowered as rapidly as possible to minus 30° F (minus 34° C). This temperature shall be maintained for a period of not less than 5 hours. The case shall then be removed from the test chamber and immediately exposed to room temperature for not less than 4 hours and examined. Evidence of deterioration or change in form or shape of any metal or wood parts; difficulty in raising or lowering of the platen carriage assembly, or in the movement of other parts; malfunction of the electrical circuitry; or signs of peeling, blistering, discoloration, or fading of any surface treatment or painting of any part or component shall constitute failure of this test.

4.5.2.1.2 High temperature. The case with tracing table, repair parts, and accessories within the case, shall be placed in a test chamber at 72° F (22° C). Temperature in the chamber shall be raised as rapidly

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as possible to 160° F (71° C) and maintained for a period of not less than 1 hour. The case and equipment shall then be removed from the test chamber and immediately exposed to room temperature of 72° F (22° C), and examined. Evidence of deterioration or change in form or shape of any metal or wood parts; difficulty in raising or lowering of the platen carriage assembly, or in the movement of other parts; malfunction of the electrical circuitry; or signs of formation of fungi, peeling, blistering, discoloration, or fading of any surface treatment or painting of any part or component shall constitute failure of this test.

4.5.2.2 Platen and platen extension. An optical microscope with a reticle graduation in 0.1 mm shall be mounted perpendicular to and independent of the platen. With the graduated reticle centered on the sapphire tip, the platen shall be rotated 360 degrees and the displacement in any direction during the rotation shall be measured. This test shall be repeated with the platen mounted on the platen extension. Evidence of displacement more than 0.1 mm in any direction shall constitute failure of this test.

4.5.2.3 Counter. The platen shall be set at its lowest range and the counter set at zero. A height gage conforming to GGG-C-111, Type IV, Class 2, and a dial indicator conforming to MIL-I-18422 shall be used to make an initial measurement of the platen height. The platen shall then be raised to a height of 100 mm as indicated on the counter. The height gage and dial indicator shall then be raised, and a second measurement shall be made on the platen. The difference between the initial and second measurement is the vertical displacement of the platen. This difference shall be checked against the counter reading. At the completion of this test, the platen height shall then be checked in not less than four more positions at the 1/4, 1/2, 3/4, and highest point of the platen's range. At each position the height of the platen shall be measured with the height gage and dial, before and after the counter has been set at zero, and then returned to the position being checked. This is to determine whether setting the counter to zero and then back to the height being checked would alter the height of the platen. A displacement of more than 0.05 mm of the platen position as indicated by the height gage shall constitute failure of this test.

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4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance wiht MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

120.	Materials, methods, or containers not as specified. Each incorrect material, method, or container shall be considered one defect.
121.	Table, repair parts, accessories, and publications not placed, cushioned, and secured within the case.
122.	Gross weight exceeds 200 pounds (90.72 kg).
123.	Strapping not as specified.
124.	Closure of the outer container for Level A not Method V as specified.
125.	Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each tracing table with repair parts, accessories, and technical publications, placed, cushioned, and secured within the case, shall be preserved in accordance with MIL-P-116, Method IIb. The case shall be considered the inner container, and the outer container shall be a close-fitting box conforming to PPP-B-636, W5c, style optional. The outer container shall be closed in accordance with the appendix to the box specification Method V.

5.1.2 Commercial. Each tracing table with repair parts, accessories, and technical publications, placed, cushioned, and secured within the case, shall be preserved in accordance with MIL-STD-1188.

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5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. The tracing tables, preserved as specified in 5.1, shall be packed in quantities not to exceed 200 pounds (90.72 kg) in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. The tracing tables, preserved as specified in 5.1, shall be packed as specified in 5.2.1 for Level A except that boxes shall be Domestic Type and strapping shall be Finish A.

5.2.3 Commercial. The tracing tables, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The tracing table, together with other multiplex equipment, is used for tracing out details and contours of the spatial model formed by projection of a properly oriented stereoscopic pair of diapositive plates in multiplex projectors.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the first produced tracing tables and number of units required (see 3.2).
(c) When color is other than the contractor's option (see 3.15 and 3.17).
(d) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(e) When Finish A strapping is required (see 5.2.1).

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6.3 First produced tracing table with case. Any changes or deviations of production tracing table(s) with case(s) from the approved first produced tracing table(s) with case(s) during production will be subject to the approval of the contracting officer. Approval of the first produced tracing table(s) with case(s) will not relieve the contractor of his obligation to furnish tracing tables with cases conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each tracing table with case.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools, as necessary (including any special tools), instructions regarding shipment of tracing table(s) with case(s).

6.6 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

Custodians:
Army - ME
Navy - AS
Air Force - 99

Review activity:
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DOCUMENT IDENTIFIER AND TITLE: MIL-T-10540D Tracing Table, Stereoplotter, Projection: with Case

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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